2026-HS1-COS10005-Web Development

Workshop Guide

Lab 6 – CSS Responsive Layout and Accessibility

Aims:

- To learn how to specify and apply different CSS rules for different devices, e.g., desktops and mobile phones;
- To get familiar with the Web Accessibility Initiative;
- To learn how to test web pages for compliance and improve the quality of web pages, and introduce some assistive technologies.

Task 1: CSS Responsive Layout (10 Marks)

We are going to create two CSS styles for one registration form, One will be used for a desktop device and the other for a mobile phone.

Step 1:

1.1 Create a new folder 'lab06' under the unit folder on the mercury server. Upload today's work to this lab06 folder.

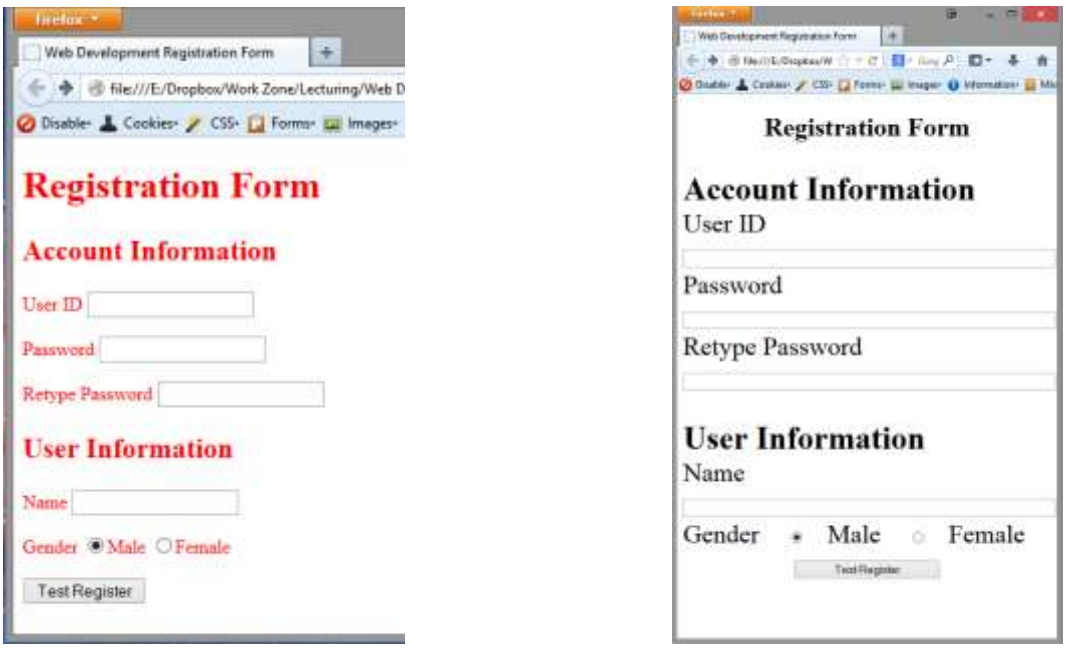
Step 2:

2.1 Download and open the text file `regform.html`.

2.2 Use VS Code to open and edit `regform.html`. Changes will need to be made in order to allow for CSS styling.

Step 3:

3.1 Design a mock up on what the form should look like in a desktop and a mobile phone. Figure 1 presents a possible mock up for both desktop and mobile phone.



a) Desktop Version **b) Mobile Version**

Figure 1: Sample Form Layouts

Step 4:

4.1 Using VS Code, create two CSS files, `desktop.css` and `mobile.css`, that will present the forms shown in Figure 1.

4.2 To apply the two CSS files to `regform.html`, you need to add the following code to the **<head>** part of `regform.html`.

```
<link href="desktop.css" rel="stylesheet" media="screen and (min-width:768px)" />
<link href="mobile.css" rel="stylesheet" media="screen and (max-width:480px)" />
```

By doing so, `desktop.css` will be applied to the `regform.html` when the width of the browser displaying `regform.html` is greater than 768 pixels, which is the usually the minimum width of a PC monitor. If the width of the browser is lower than 480 pixels, `mobile.css` will be applied to `regform.html`, displaying the web page in a mobile-friendly manner, e.g., larger font and compact layout.

Step 5:

5.1 Specify only one CSS rule in `desktop.css`:

```
body {
  color:red;
}
```

Using this CSS rule, when all the text on the web page are rendered red (see Figure 1a), we know *`desktop.css`* is taking effect, not *`mobile.css`*.

5.2 Open *`mobile.css`*. Following the comments below, complete and apply CSS rules to present *`regform.html`* as shown in Figure 1b. You might need to specify the `id` or the `class` attributes of applicable HTML elements so that they can be properly selected in the CSS file for CSS application.

[IMPORTANT] Complete one CSS rule at a time and test the webpage to understand the effect of that CSS CSS rule. If you cannot see a difference when testing your webpage, try reducing the width of your browser window.

When does `mobile.css` kick in?

```
______ {
  ________:__________;        /* Increase the font size in the form to 200% */
}

{
  ________:__________;        /* Center the text for all <h1> elements */
}

______ {
  ________:__________;        /* Change the font size of <h2> to 40 pixels */
  ________:__________;        /* Remove the bottom margin of <h2> */
```

```
}

______ {
  ________:__________;  /* Make all the <input> elements occupy
                           the entire width  */
}

______ {
  ________:__________;        /* Remove all margins around all <p>
elements */
}

______  {
  ________:__________;        /* Set width of all radio buttons to
15% */
}

______ {
  ________:__________;        /* Set width of all buttons to 40%
*/
}

______ {
  ________:__________;        /*Center the test button by
centering the appropriate <div> */
}
```

Step 6:

Merge `desktop.css` and `mobile.css` into one CSS file named `style.css`.

6.1 Create a file named `style.css`.

6.2 Add the following CSS rule to `style.css`:

```
@media screen and (min-width: 1024px) {
    body {
        font-size: 100%;
    }
}
```

6.3 Apply the media query below to all your code from **<u>Step 5.2</u>**:

```
@media screen and (max-width: 480px) {
    …
    …
}
```

6.4 Remove the following code from `regform.html`.

```
<link href="desktop.css" rel="stylesheet" media="screen and (min-
                         width:768px)" />
<link href="mobile.css" rel="stylesheet" media="screen and (max-
                         width:480px)" />
```

6.5 Add the following code to the **<head>** part of `regform.html`.

```
<link href="style.css" rel="stylesheet" />
```

Now only one CSS file is applied to `regform.html`. Based on the width of the browser window, different CSS rules will kick in and present `regform.html` in two different ways. Test it and see for yourselves.

Step 7. Test and view web pages.

7.1 Using WinSCP (or Cyberduck for Mac users), upload your files, including `regform.html`, `desktop.css`, `mobile.css` and `style.css` onto Mercury.

7.2 Now you can even use your mobile phone to test your webpages.

7.3 To view the pages through http, use any Web browser and type in the following address,

http://mercury.swin.edu.au/<your unit code>/***s***<*your Swinburne ID*>/<folder>/<filename>

Please refer to the following examples to identify the URLs of your web pages.

Folder on Mercury Web Server	**URL**
~/cos10005/www/htdocs/index.html	http://mercury.swin.edu.au/cos10005/s1234567/index.html
~/cos60002/www/htdocs/lab06/regform.html	http://mercury.swin.edu.au/cos60002/s1234567/lab06/regform.html

Note: You can copy the URLs in the table, but remember to replace the unit codes and student id in the above examples with yours to obtain the URLs of your web pages on Mercury.

[IMPORTANT] When the browser authorization request dialog pops up, use your SIMS username *and* password to confirm access.

Step 8: HTML and CSS Validation

To validate the HTML file use the validator at http://validator.w3.org.

To validate the CSS file use the CSS validator at http://jigsaw.w3.org/css-validator/